DAY11-KALI-SETUP

THEORY (answer all 20 questions in your own words):

1. What is Kali Linux?
   Debian based linux distribution specifically designed for cybersecurity, security testing, pentration testing, digital forensics, security research and CTFs.

2. Why is Kali Linux used in cybersecurity?
   Kali Linux is used in cybersecurity cause it learns you learn without replacing your main OS. Mostly used by cybersecurity professionals to learn and get hands on pratice with it.
3. What is a virtual machine?
   It's a software based computer running in it's own OS and application separately on a physical host machine.

4. What is a hypervisor?
   It is the main OS on the physical computer
   Eg: Windows 11, Linux, MAC OS.

5. What is VirtualBox?
   Virtualization software used to create and run VMs such as Kali Linux, Ubuntu, Windows etc…

6. Explain Host OS and Guest OS.
   Host OS: Main OS on Physical computer.
   Eg: windows 11, Linux.
   Guest OS: Runs inside Virtual machine.
   Eg: Windows Host, virtual box.

7. Why is using Kali inside a VM useful for beginners?
   Because it comes along with a lot of tools that are beginner oriented and gives hands on training just not theory and industry based.

8. What is a VM snapshot?
   A VM snapshot is taking a picture of the current state of the VM and using it as a recoverable state rather than reinstalling the whole thing.

9. Why should students create snapshots?
   Helpful for recovery, if incase the VM gets crashed or data is lost.

10. What does whoami display: Username

11. What does hostname display: Name assigned to the computer

12. What does uname -a display: Linux kernel and system information.

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13. What does /etc/os-release contain: About linux OS such as name and version.

14. What does ip addr show: Network Interface and IP configuration.

15. What does ip route show: Routing Information, also default route.

16. What is sudo?
    Sudo is a command-line tool on operating systems such as Linux and macOS that lets a user run a program with the security privileges of another user, typically the superuser or root.

17. Why should elevated privileges be used carefully?
    Because Unauthorized user getting privilege can make data lost, identity theft, spamming or ransomware etc..

18. What does sudo apt update do: Refresh local packet respoitory information.

19. Why should Kali Linux tools only be used on authorized systems?

    Kali contains powerful tools. The same tools used by security engineers, penetration testers, researchers and red teams can be misused. So authorized sytem is necessity.

20. Explain the difference between ethical security testing and unauthorized access.
    Ethical Security Testing: Performed with permission.
    Unauthorized access: Without permission

DAY 11 PRACTICAL LAB — Build and Explore Your Kali Linux Lab

LAB OPTION A: if you already have a Kali VM — use it; do NOT reinstall.

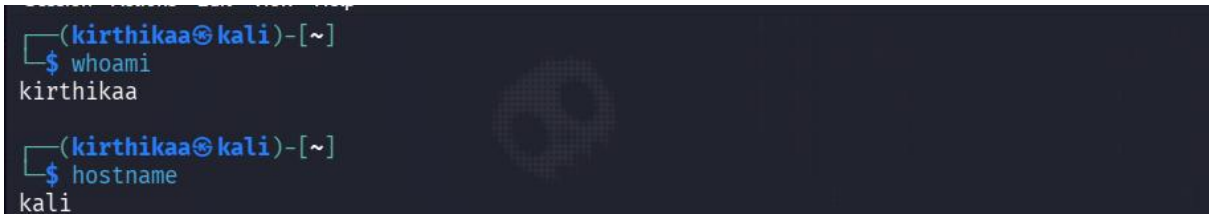
LAB OPTION B: if you don't — install/import Kali in VirtualBox from the OFFICIAL Kali images only. Never use or share unofficial modified images.

TASK 1 — Start the Kali VM → confirm the desktop loads.

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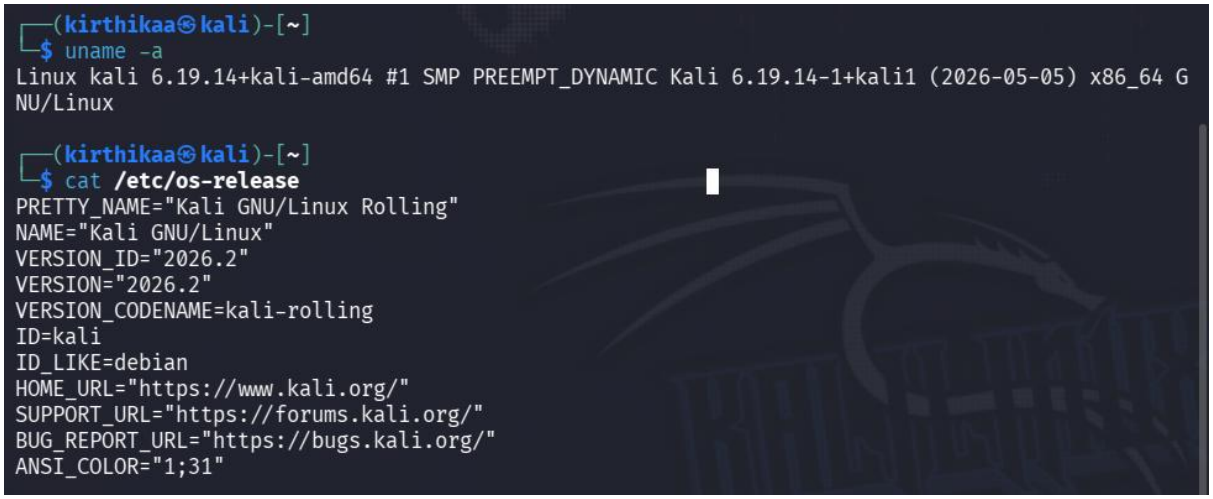
TASK 2 — Terminal: whoami and hostname → record Username and Hostname.



TASK 3 — pwd and ls → record the current directory.



TASK 4 — uname -a then cat /etc/os-release → identify Operating System and Kernel.



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TASK 5 — ip addr then ip route → identify Primary Interface, IPv4 Address, Default Route. (Keep this private — never publish your network info.)

```
┌──(kirthikaa㉿kali)-[~]
└─$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
1000
    link/ether 08:00:27:af:ed:06 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 78404sec preferred_lft 78404sec
    inet6 fd17:625c:f037:2:67f2:346d:943a:b7f2/64 scope global temporary dynamic
       valid_lft 86334sec preferred_lft 14334sec
    inet6 fd17:625c:f037:2:a00:27ff:feaf:ed06/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86334sec preferred_lft 14334sec
    inet6 fe80::a00:27ff:feaf:ed06/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(kirthikaa㉿kali)-[~]
└─$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
```

TASK 6 — ping -c 4 google.com → record whether connectivity works.

```
┌──(kirthikaa㉿kali)-[~]
└─$ ping -c 4 google.com
PING google.com (172.217.25.78) 56(84) bytes of data.

── google.com ping statistics ──
4 packets transmitted, 0 received, 100% packet loss, time 3066ms
```

TASK 7 — sudo apt update → observe the repository information refresh. No full upgrade needed.

```
┌──(kirthikaa㉿kali)-[~]
└─$ sudo apt update
Get:1 https://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 https://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.1 MB]
Get:4 https://kali.download/kali kali-rolling/contrib amd64 Packages [104 kB]
Get:5 https://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [184 kB]
Get:6 https://kali.download/kali kali-rolling/non-free amd64 Packages [168 kB]
Get:7 https://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [886 kB]
Get:8 https://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.3 kB]
```

TASK 8 — cd ~ · mkdir jayvik-cyber-lab · cd jayvik-cyber-lab · pwd · touch day11-notes.txt · ls -l

```
┌──(kirthikaa㉿kali)-[~]
└─$ cd ~

┌──(kirthikaa㉿kali)-[~]
└─$ mkdir jayvik-cyber-lab

┌──(kirthikaa㉿kali)-[~]
└─$ cd jayvik-cyber-lab

┌──(kirthikaa㉿kali)-[~/jayvik-cyber-lab]
└─$ pwd
/home/kirthikaa/jayvik-cyber-lab

┌──(kirthikaa㉿kali)-[~/jayvik-cyber-lab]
└─$ touch day11-notes.txt

┌──(kirthikaa㉿kali)-[~/jayvik-cyber-lab]
└─$ ls -l
total 0
-rw-rw-r-- 1 kirthikaa kirthikaa 0 Oct 15 16:43 day11-notes.txt
```

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TASK 9 — Create a VirtualBox snapshot named Day11-Clean-Kali and understand how to restore it. (Do NOT break the VM to test restore.)

